

PROJECT PROPOSAL

on

**Predictive QoS Preemption Engine: An ML-Driven 5G Network
Slice Controller for Emergency First Responder Communications**

By

Team ZeroLag

From

SNS College of Technology

1. Objective

The primary objective of this project is to design, develop, and validate a Predictive Quality of Service (QoS) Preemption Engine — a software-defined, machine learning (ML)-powered network controller for 5G infrastructure that proactively manages cellular congestion during localized emergencies. The engine will monitor real-time network telemetry and multi-source environmental signals, forecast impending cell congestion before it occurs, and autonomously provision an ultra-reliable, low-latency (URLLC) network slice exclusively dedicated to authenticated emergency services — shifting the network management paradigm from reactive to proactive.

Specifically, the proposed work aims to:

- Develop a multi-source ML prediction model — integrating live radio access network (RAN) telemetry, Emergency 911 (E911) call spike data, social media velocity signals, IoT sensor anomalies, and event calendar feeds — to forecast cellular congestion at least 45–90 seconds in advance with greater than 90% accuracy.
- Design a graduated civilian traffic throttling framework that avoids abrupt service termination and instead applies tiered bandwidth reductions (video quality degradation before session drops) to preserve both civilian service continuity and emergency bandwidth.
- Engineer a role-based, authenticated URLLC slice sub-partitioning mechanism that maps different categories of emergency devices (Mission Critical Push-to-Talk voice, telemetry uplinks, drone video streams) to differentiated SLA sub-slices with guaranteed latency and throughput.
- Integrate an Explainable AI (XAI) dashboard enabling network operations center (NOC) personnel to audit, understand, and override engine decisions, providing human-in-the-loop safety for life-critical operations.
- Validate the complete system through digital twin simulation and controlled testbed experiments, demonstrating measurable improvements in emergency service latency, throughput, and network reliability over existing reactive QoS schemes.

2. Summary of Proposal

During large-scale localized emergencies — including natural disasters, mass-casualty incidents, and civil crises — civilian populations instinctively reach for their smartphones, generating sudden and massive spikes in cellular data traffic. These surges routinely overwhelm local 5G cell sites, depriving first responders of the critical network bandwidth required for life-saving telemetry, body-worn camera feeds, Mission Critical Push-to-Talk (MCPTT) voice coordination, and drone-assisted situational awareness. Contemporary solutions such as AT&T FirstNet, T-Mobile T-Priority, and Verizon FrontLine address this

problem through reactive priority and preemption mechanisms — they elevate first responder traffic after congestion has already begun. No commercially deployed system today predicts and prevents congestion before a network performance collapse occurs.

This proposal presents the Predictive QoS Preemption Engine (PQPE), a novel ML-driven software controller deployed within the Open Radio Access Network (O-RAN) Near-Real-Time RAN Intelligent Controller (Near-RT RIC) layer. PQPE continuously ingests live network KPIs alongside heterogeneous external signals to train and operate a multi-source congestion prediction model. Upon detecting a high-confidence congestion forecast, PQPE autonomously executes a pre-defined preemption cascade: first applying graduated civilian throttling to reduce non-essential data consumption, then forcefully provisioning a dedicated URLLC network slice exclusively for authenticated emergency services — all before network performance degrades below first responder operational thresholds.

The proposed engine introduces several innovations over the existing state of the art: (i) multi-modal congestion prediction combining network telemetry with social, geographic, and event-based signals; (ii) tiered, graduated civilian throttling rather than abrupt session preemption; (iii) role-aware emergency slice sub-partitioning with SLA differentiation per device class; (iv) a federated edge inference architecture enabling sub-10 millisecond decision latency; and (v) a digital twin pre-validation layer and XAI dashboard for transparent, auditable decision-making. The system is designed to integrate natively with 3GPP Release 17/18 5G Standalone (SA) architectures and O-RAN xApp frameworks, ensuring compatibility with current and near-future commercial 5G infrastructure. The project is proposed for a duration of three years with a total estimated budget of Rs. 45 Lakhs.

3. Detailed Project Proposal

3.1 Background and Problem Statement

The integration of 5G into public safety infrastructure represents one of the most consequential technological transitions of the current decade. First responders increasingly rely on bandwidth-intensive applications including real-time HD video streaming from body-worn cameras and unmanned aerial vehicles (UAVs), GIS-based situational awareness mapping, remote medical telemetry, and MCPTT group communications. The reliable operation of all these systems is contingent upon guaranteed Quality of Service (QoS) parameters — specifically ultra-low latency (below 5 ms for URLLC applications), high throughput (up to 5 Gbps per cell for video), and near-100% availability.

Contemporary emergency network prioritization systems, while operationally valuable, function reactively. T-Mobile T-Priority and Verizon FrontLine execute preemption only after a first responder device connects and requests elevated priority, meaning the preemption action is triggered after congestion has already

begun competing for resources. AT&T FirstNet, operating on dedicated Band 14 spectrum, provides greater isolation but still lacks a proactive, ML-driven congestion forecast layer that can anticipate and pre-position network resources before an incident-driven traffic surge materializes. Academic research — including the RESPOND-A platform (MDPI Energies, 2021) and the IEEE proactive uplink slicing framework (IEEE Xplore DOI: 10.1109/10001244) — has addressed individual components of this problem but no integrated, deployable system combining multi-modal prediction, graduated civilian throttling, and role-aware slice provisioning has been demonstrated.

3.2 Proposed System Architecture

The PQPE is architected as an O-RAN xApp deployed within the Near-RT RIC (10 ms – 1 second control loop) with a companion rApp in the Non-RT RIC (greater than 1 second control loop) for model training and policy management. The system comprises five tightly integrated modules:

3.2.1 Multi-Source Congestion Prediction Module

The prediction module implements a hybrid deep learning architecture combining Long Short-Term Memory (LSTM) networks for sequential RAN telemetry processing with a Graph Neural Network (GNN) layer to model spatial inter-cell dependency and congestion propagation across cell clusters. The model ingests the following input signal streams:

- Live RAN KPIs: Physical Resource Block (PRB) utilization per cell, uplink/downlink throughput, SINR distributions, active UE count, and handover rates — sampled at 100 ms intervals via the O-RAN E2 interface.
- Emergency Trigger Signals: E911 call volume per PSAP (Public Safety Answering Point) catchment area, updated every 10 seconds via API integration with NG-911 systems.
- Social Media Velocity: Real-time monitoring of geotagged post density and keyword spike rates (disaster, fire, accident, etc.) on public APIs, processed through a BERT-based NLP classifier.
- IoT Environmental Sensors: Anomalous readings from connected infrastructure sensors (smoke detectors, traffic monitoring nodes, seismic sensors) within the cell coverage zone.
- Event Calendar Integration: Pre-loaded data on scheduled mass gatherings, sporting events, and civic events, enabling pre-incident resource pre-positioning.

The model targets a minimum 45-second prediction horizon with greater than 90% precision and less than 5% false positive rate, validated against historical incident datasets from PSAP records and network operator CDR databases.

3.2.2 Graduated Civilian Throttling Module

Upon a high-confidence congestion forecast, the engine executes a pre-defined, staged throttle cascade targeting non-essential civilian data streams in descending order of bandwidth consumption. The cascade proceeds through the following tiers, with each tier activated progressively only if the preceding tier does not sufficiently reduce predicted load:

- Tier 1 — Video Quality Reduction: Cap 4K and Ultra HD streaming to 1080p via dynamic QoS class identifier (QCI) downgrade for eMBB slices serving consumer video platforms.
- Tier 2 — Upload Rate Limiting: Restrict civilian uplink throughput for social media upload and cloud backup applications, which contribute disproportionately to uplink congestion during emergency events.
- Tier 3 — Background Application Throttling: Suppress throughput to zero for background application traffic categories (software updates, cloud sync) while maintaining foreground user-facing traffic.
- Tier 4 — Session Preemption (Last Resort): Drop non-priority sessions in strict compliance with 3GPP TS 23.501 preemption vulnerability and preemption capability parameters, prioritizing shorter-duration drops over permanent disconnections.

This graduated approach is both technically superior (minimizing unnecessary civilian service degradation) and operationally critical (reducing public and regulatory backlash against emergency preemption mechanisms).

3.2.3 Role-Aware URLLC Slice Provisioning Module

Concurrent with civilian throttling, the engine provisions a dedicated emergency URLLC network slice through the 5G Service Management and Orchestration (SMO) framework and NFVO. Emergency devices are authenticated via digital certificate chains anchored to agency-issued credentials, and mapped to role-specific sub-slices with differentiated SLA parameters:

Device Class	Application	Target Latency	Min. Throughput
Command & Control Vehicle	Drone Video / Situational Awareness	< 10 ms	50 Mbps
Paramedic / EMS Device	Vital Signs Telemetry / HL7 Streams	< 5 ms (URLLC)	5 Mbps
Firefighter Device	MCPTT Voice / Location Beacon	< 300 ms	2 Mbps
Incident Commander Tablet	GIS Maps / Body Cam Streams	< 20 ms	25 Mbps
Drone (UAV)	4K Video Uplink	< 15 ms	40 Mbps

3.2.4 Digital Twin Pre-Validation Layer

Before any preemption action is committed to the live network, the engine submits the proposed intervention to a real-time digital twin replica of the affected cell cluster. The digital twin, maintained as a continuously synchronized shadow model using live network state snapshots, simulates the expected outcome of the proposed throttling and slicing actions — including potential effects on adjacent cells — within 500 milliseconds. Only interventions that the digital twin validates as achieving the target emergency QoS improvement without causing unintended cascading congestion in neighbouring cells are committed to execution. This layer is a critical safety mechanism for life-critical network operations.

3.2.5 Explainable AI Dashboard

The XAI dashboard provides NOC personnel with real-time, human-readable explanations of each engine decision through SHAP (SHapley Additive exPlanations) value decomposition, displaying the contribution weight of each input signal to the congestion prediction score. The dashboard provides: a live congestion forecast heatmap overlaid on the geographic cell topology; the specific throttling tier activated and civilian segments affected; the emergency slice status, authenticated device count, and per-device SLA achievement metrics; and a one-click manual override capability for trained operators. All engine decisions, predictions, and overrides are logged to an immutable audit trail for post-incident review and regulatory compliance.

3.3 Implementation Methodology

The project will be executed in four phases across a three-year duration:

- Phase 1 (Months 1–9): Literature consolidation, dataset acquisition (PSAP incident records, network CDR data, social media archives), O-RAN testbed setup using OpenAirInterface (OAI) or ORAN-SC (O-RAN Software Community), and development of the baseline LSTM congestion prediction model.
- Phase 2 (Months 10–18): Multi-modal prediction model development integrating GNN spatial layers, social signal NLP classifier, and IoT sensor fusion. Training, validation, and benchmarking against the reactive preemption baseline. Development of the graduated throttling policy engine.
- Phase 3 (Months 19–27): Implementation of the role-aware URLLC slice provisioning module, PKI-based device authentication framework, digital twin simulation integration, and full system integration within O-RAN Near-RT RIC as an xApp.
- Phase 4 (Months 28–36): System-level testing on physical 5G SA testbed, performance validation under simulated emergency traffic scenarios, XAI dashboard development and usability evaluation, documentation, publication of results, and technology transfer preparation.

3.4 Work Schedule (BAR Diagram Summary)

Activity	M1-9	M10-18	M19-27	M28-36
Literature survey & dataset acquisition	✓			
Baseline LSTM prediction model	✓			
Multi-modal GNN + NLP fusion model		✓		
Graduated throttling policy engine		✓		
URLLC slice provisioning & device auth.			✓	
Digital twin integration			✓	
Full O-RAN xApp integration			✓	
Testbed validation & performance testing				✓
XAI dashboard & audit framework				✓
Publications, documentation, TT				✓

4. Societal & Industrial Impact

4.1 Societal Impact

The most direct societal impact of the PQPE is the preservation of human life during emergencies. Network congestion-induced communication failures during disasters are documented contributors to delayed emergency response, failed medical interventions, and coordination breakdowns among multi-agency response teams. By ensuring that first responders maintain uninterrupted, guaranteed-QoS connectivity during the most critical phase of any emergency — the first 60 minutes — the proposed system directly reduces preventable mortality and injury. The system's graduated civilian throttling approach is also designed to minimize collateral civilian communication disruption: social connectivity during emergencies is itself a public safety asset, enabling families to confirm safety, enabling bystander reporting, and reducing panic-driven information voids.

Beyond direct emergency response, the PQPE framework is directly applicable to mass-gathering event management (festivals, stadium events, political rallies), infrastructure monitoring, and smart city resilience planning. The XAI audit trail additionally supports democratic accountability by providing transparent records of when, why, and how civilian communications were prioritized or throttled — a critical safeguard against misuse of preemption authority.

4.2 Industrial Impact

The PQPE addresses a well-defined gap in the commercial 5G network management stack. Telecom operators investing in emergency services contracts — including national initiatives analogous to FirstNet in India, the UK Emergency Services Network (ESN), and South Korea's PS-LTE program — require differentiated, proactive network intelligence to meet Government SLA commitments. The proposed system can be offered as a value-added xApp module by RAN vendors, system integrators, and software-defined networking platform providers operating in the O-RAN ecosystem.

Industrial beneficiaries include:

- 5G infrastructure vendors and managed service providers seeking differentiated emergency-grade network intelligence products.
- Public Safety Answering Points (PSAPs) and National Disaster Management Authorities (NDMAs) requiring reliable connectivity SLA guarantees for first responder fleets.
- Smart city operators integrating 5G infrastructure with emergency response planning frameworks.
- Defence and paramilitary organizations requiring URLLC tactical communication slices in contested or high-congestion environments.

The modular, standards-compliant O-RAN xApp architecture ensures that the PQPE can be commercially deployed without requiring proprietary hardware modifications, significantly reducing time-to-market and integration cost for adopters.

5. Scalability & Market Readiness

5.1 Technical Scalability

The PQPE is architected from the ground up for horizontal scalability. The Near-RT RIC xApp framework allows the engine to be deployed as containerized microservices (Kubernetes-orchestrated), enabling elastic scaling across multiple cell clusters, regions, and network operators without re-architecture. The federated ML inference model is designed for edge deployment at the Distributed Unit (DU) or Centralized Unit (CU) level, ensuring that prediction latency remains below 10 milliseconds regardless of the number of cells being monitored, by distributing inference across the O-Cloud edge compute fabric.

Key scalability features include:

- Hierarchical control loops: Non-RT RIC rApp handles model retraining and policy updates (> 1 second loop) while Near-RT RIC xApp handles real-time preemption decisions (10 ms – 1 second loop), ensuring that computational load is distributed appropriately across the O-RAN hierarchy.

- Federated learning architecture: Cell-site edge nodes train local congestion prediction sub-models on local traffic data without centralizing raw sensitive traffic records, enabling privacy-preserving scaling across thousands of cell sites.
- Stateless API design: All PQPE modules expose RESTful APIs compliant with O-RAN Alliance A1 and E2 interface specifications, ensuring interoperability with any 3GPP Release 17/18-compliant 5G SA core and RAN infrastructure.
- Multi-operator interoperability: The authentication and slice provisioning modules support cross-operator roaming credentials, enabling first responders to seamlessly access emergency slices on any participating network — critical for large-scale disaster scenarios exceeding a single operator's coverage footprint.

5.2 Market Readiness

The global public safety LTE and 5G market is a high-growth segment, driven by government mandates and national emergency broadband programs worldwide. The PQPE targets a positioning as a software product within the O-RAN xApp marketplace — a rapidly growing commercial ecosystem — with an anticipated go-to-market pathway within 18–24 months post-project completion through partnerships with O-RAN Alliance members and telecom operators engaged in public safety contracts.

Readiness Dimension	Current Status	End-of-Project Target
Prediction model (LSTM/GNN)	Research prototype (TRL 3)	Validated xApp (TRL 6)
Civilian throttling engine	Conceptual design (TRL 2)	Testbed-validated (TRL 5)
URLLC slice provisioning	Standards-defined (TRL 4)	Live testbed (TRL 6)
Digital twin integration	Proof of concept (TRL 3)	Integrated system (TRL 5)
XAI dashboard	Conceptual (TRL 2)	User-validated prototype (TRL 5)
O-RAN xApp packaging	Not started (TRL 1)	Production-ready package (TRL 6)

6. 5G Technology Utilization

6.1 Core 5G Technologies Exploited

The PQPE is deeply embedded in the 5G technology stack and exploits the following 5G-native capabilities that have no equivalent in LTE or prior generations:

6.1.1 5G Network Slicing (3GPP TS 23.501)

Network slicing — the ability to create multiple logically independent virtual networks on a single physical 5G infrastructure — is the foundational 5G technology that makes the PQPE's emergency slice provisioning concept technically feasible. Unlike LTE priority and preemption (which operates at the session/bearer

level), 5G network slicing allows the PQPE to provision a fully isolated network instance with its own dedicated resources, routing policies, and QoS guarantees. The engine specifically leverages URLLC slice type (SST=2 in 3GPP nomenclature) with customized slice-specific parameters for latency, reliability, and packet loss targets, differentiated from the eMBB slice (SST=1) serving civilian traffic.

6.1.2 O-RAN Open Architecture (O-RAN Alliance Specifications)

The O-RAN disaggregated RAN architecture provides the PQPE's deployment home. The Near-RT RIC (xApp) and Non-RT RIC (rApp) framework, standardized by the O-RAN Alliance, creates a vendor-neutral, programmable intelligence layer over the RAN — enabling the PQPE to ingest standardized KPM (Key Performance Metric) reports via the E2 interface, issue policy updates via the A1 interface, and control resource allocation via the O1 interface — all without dependence on proprietary vendor APIs. This open architecture is essential to the PQPE's market scalability across multi-vendor deployments.

6.1.3 5G Standalone Core (5GC) — 3GPP Releases 15–18

The 5G Standalone core introduces the Service-Based Architecture (SBA) that exposes all core network functions as microservices with standardized APIs. The PQPE integrates with the Network Slice Selection Function (NSSF) for slice selection, the Policy Control Function (PCF) for QoS policy enforcement, the Session Management Function (SMF) for dynamic session modification during throttling, and the Network Data Analytics Function (NWDAF) for real-time network analytics data feeds. NWDAF in particular is a 5GC-native ML analytics function that the PQPE leverages as a complementary data source, feeding into the multi-modal congestion prediction model.

6.1.4 Multi-Access Edge Computing (MEC)

The engine's federated inference architecture is deployed on 5G MEC nodes co-located with the 5G Distributed Unit (DU) or Central Unit (CU), enabling sub-10 millisecond prediction-to-action latency. MEC reduces round-trip dependence on the central cloud, which is especially critical during large-scale disasters where backhaul links may themselves be impaired. The digital twin simulation layer is also hosted at the MEC layer, enabling real-time pre-validation of preemption decisions with minimal additional latency.

6.1.5 5G QoS Framework — QCI / 5QI Parameters

The 5G QoS Identifier (5QI) framework provides the granular, standardized traffic classification mechanism that the graduated throttling module operates on. The PQPE dynamically reassigns 5QI values to civilian traffic flows — reducing video streams from 5QI 2 (conversational video, 150 ms PDB) to 5QI 4 (non-conversational video), then further suppressing background traffic classified as 5QI 8/9 — in a controlled, standards-compliant manner that preserves interoperability with all 3GPP-compliant UEs and network functions.

6.2 Alignment with National 5G Policy

India's National Telecom Policy and the Department of Telecommunications' 5G rollout framework specifically identify public safety communications and smart city emergency management as priority use cases for 5G deployment. The PQPE directly addresses these policy priorities by delivering a deployable, standards-based solution that enables Indian telecom operators to meet regulatory obligations for emergency priority access without requiring dedicated public safety spectrum investments analogous to FirstNet's Band 14 model. This positions the PQPE as a cost-effective, spectrum-efficient complement to India's commercial 5G infrastructure — leveraging existing 5G SA deployments on commercial bands to serve national public safety objectives.

7. Proposed Budget

Expenditure Head	Year 1 (Rs. Lakhs)	Year 2 (Rs. Lakhs)	Year 3 (Rs. Lakhs)	Total (Rs. Lakhs)
Staff Salaries (5 JRF + 1 Project Associate)	13.20	13.20	13.20	39.60
Computing Equipment (GPU Workstations, Edge Server)	5.0	1.0	0.0	6.0
O-RAN Testbed Components & SDRs	4.0	2.0	0.5	6.5
Software Licenses & Cloud Compute (AWS/Azure)	1.5	1.5	1.0	4.0
Travel (Conferences, Lab Visits)	0.5	0.75	0.75	2.0
Contingencies & Consumables	1.0	1.0	1.0	3.0
Institutional Overheads @ 20%	5.04	3.89	3.29	12.22
TOTAL	30.24	23.34	19.74	73.32

8. Key References

The following peer-reviewed works form the primary literature foundation for this proposal:

- [1] "Proactive Resource Management for Predictive 5G Uplink Slicing," IEEE Xplore, DOI: 10.1109/10001244, 2022.
- [2] "Admission and Congestion Control for 5G Network Slicing," IEEE Conference Publication, DOI: 10.1109/8581773, 2018.
- [3] Gardikis et al., "Conceptual Evaluation of a 5G Network Slicing Technique for Emergency Communications," MDPI Energies, vol. 14, no. 21, 2021.
- [4] Carlberg, Burger, Jover, "Dynamic 5G Network Slicing for First Responders," Proc. IPTComm, Chicago, USA, 2019.
- [5] Perez-Romero et al., "Supporting Mission Critical Services through RAN Slicing," Proc. ICT-DM, Paris, France, 2019.
- [6] Alnawayseh et al., "Smart Congestion Control in 5G/6G Networks Using Hybrid Deep Learning Techniques," Wiley Complexity, 2022.
- [7] "AI-Driven 5G Network Optimization: A Comprehensive Review," American Journal of Artificial Intelligence, Science Publishing Group, 2024.
- [8] "AI-Driven 5G/6G-Satellite Hybrid Networks: A Real-Time Framework for Emergency Decision-Making," IEEE ComSoc Young Professionals, February 2026.
- [9] 3GPP TS 23.501: System Architecture for 5G System (Release 17), 3rd Generation Partnership Project, 2022.
- [10] O-RAN Alliance, "O-RAN Near-Real-Time RAN Intelligent Controller Architecture," O-RAN WG3 Specification, 2023.